

The following MCQs and short questions cover the topic for **Elasticity**. Solve these for thorough practice and for help in any question, visit me at my consultation hours after dropping a mail at eelma.ahamed@g.bracu.ac.bd.

Multiple choice questions

1. If the elasticity of demand for college textbooks is -0.1 , and the price of textbooks increases by 20%, how much will the quantity demanded change, and in what direction?
 - a. The quantity demanded increases by 2%
 - b. The quantity demanded decreases by 20%
 - c. The quantity demanded decreases by 2%
 - d. The quantity demanded remains the same

2. If the elasticity of demand for spring break packages to Cancun is -5 , and if you notice that this year in Cancun the quantity of packages demanded increased by 10%, then what happened to the price of Cancun vacation packages?
 - a. The price fell by 10 percent
 - b. The price fell by 2 percent
 - c. The price increased by 2 percent
 - d. The price remained the same

3. In your college town, real estate developers are building thousands of new student friendly apartments close to campus. If you want to pay the lowest rent possible, should you hope that demand for apartments is elastic or inelastic?
 - a. Elastic
 - b. Inelastic

4. In your college town, the local government decrees that thousands of apartments close to campus are uninhabitable and must be torn down next semester. If you want to pay the lowest rent possible, should you hope that demand for apartments is elastic or inelastic?
 - a. Elastic

b. Inelastic

5. The long-run elasticity of oil demand has been estimated at -0.5. If the price of oil rises by 10%, how much will the quantity of oil demanded fall?
 - a. 5%
 - b. 0.5%
 - c. 2%
 - d. 20%
6. The long-run elasticity of oil demand has been estimated at -0.5. Does a 10% rise in oil prices increase or decrease total revenues to the oil producers?
 - a. Increase
 - b. Decrease
7. The elasticity of demand is 0.2. Is the demand curve relatively steep or flat? Will a fall in price raise total revenue or lower it? Note: we present the elasticity in terms of its absolute value.
 - a. Relatively steep; raise total revenue
 - b. Relatively flat; raise total revenue
 - c. Relatively steep; lower total revenue
 - d. Relatively flat; lower total revenue
8. The elasticity of demand is 2.0. Is the demand curve relatively steep or flat? Will a fall in price raise total revenue or lower it? Note: we present the elasticity in terms of its absolute value.
 - a. Relatively steep; raise total revenue
 - b. Relatively flat; raise total revenue
 - c. Relatively steep; lower total revenue
 - d. Relatively flat; lower total revenue

9. Henry Ford famously mass-produced cars at the beginning of the twentieth century, starting Ford Motor Company. He made millions because mass production made cars cheap to make, and he passed some of the savings to the consumer in the form of a low price. Cars became a common sight in the United States thereafter. Keeping total revenue and its relationship with price in mind, do you expect the demand for cars to be elastic or inelastic given the story of Henry Ford?
- a. Elastic
 - b. Inelastic
10. If the cross price elasticity between butter and peanut butter is 2, which is correct?
- a. The two goods are complements, and a price increase in one good will cause an increase in the quantity demanded of the other.
 - b. The two goods are independent, and a price increase in one good will cause an increase in the quantity demanded of the other.
 - c. The two goods are substitutes, and a price decrease in one good will cause an increase in the quantity demanded of the other.
 - d. The two goods are substitutes, and a price increase in one good will cause an increase in the quantity demanded of the other.

Short questions

1. Given the following data for the supply and demand of movie tickets, calculate the price elasticity of supply when the price changes from \$9.00 to \$10.00.

Price	Quantity Demanded	Quantity Supplied
\$7	100	25
\$8	90	45
\$9	75	75
\$10	55	105
\$11	30	125

2. If the PES is 2.0 for pen drives: and the firm supplied 4,000 when the price was £40. If the price increased from £40 to £46, what would be the new Q?
3. Yesterday the price of envelopes was \$3 a box, and Julie was willing to buy 10 boxes. Today, the price has gone up to \$3.75 a box and Julie is now willing to buy 8 boxes. What kind of demand is Julie's demand for an envelope? Explain.
4. When the price of cheese increases by 40 percent, the quantity demanded of pizza decreases by 60 percent. What is the cross-price elasticity of demand for pizza and cheese?
5. a) Find the elasticity of demand between the price of \$1 and price \$1.50 with the quantity demanded being 500 and 400 respectively, using the midpoint method.
Next, find the total revenue before and after the change in price.

b) Find the income elasticity when the income increases from \$20,000 to \$30,000 with quantity changing from 10 to 2.

Solutions

McQs

1. c
2. b
3. b
4. a
5. a
6. a
7. c
8. b
9. a
10. d

Short questions

1. $(0.3333...)/(0.1053...) = 3.167$
We get the price elasticity of supply when the price increases from \$9 to \$10 is 3.16. So for movie tickets,
The price is elastic and thus supply is very sensitive to price changes.
2. Price increases by £6 (40-46), therefore as a % $6/40 = 15\%$

PES = % change in QS/ % change in price

$$2.0 = \% \text{ change in QS} / 15$$

$$30 = \% \text{ change in QS}$$

$$\text{Therefore new Q} = 5200$$

3. PED=1 (unit elastic demand)

Quantity demand will respond proportionately to a change in price.